

# LEVERAGE

Options Auto-Execution Platform

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## Technical Documentation

System Architecture · Quantitative Models · Engineering Reference

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**Version:** 1.0 (HEAD: migration 019)  
**Deployment:** leveragequant.tech (AWS Lightsail)  
**Stack:** Python/Flask · React 18 · PostgreSQL  
**Test baseline:** 1,065 passed, 5 pre-existing failures  
**Date:** 2026-08-12

**Accuracy Policy:** All facts are verified against the live repository.  
NOT FOUND IN REPOSITORY = not found. INFERENCE = not explicitly in code.  
UNCERTAIN = requires human confirmation. Nothing is invented.

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# Chapter 1

## Executive Overview

LEVERAGE is an options analytics and paper-trading auto-execution platform for Indian index derivatives (NIFTY, BANKNIFTY, FINNIFTY). The system has two primary surfaces:

1. **Analytics Platform** — an institutional-grade options analytics dashboard covering Greeks exposure (GEX/DEX/VEX/Charm), dealer positioning, flow / microstructure, volatility surface, strategy builder, screener, and more.
2. **Paper Trading Auto-Execution Engine** — simulates options order execution driven by the signal engine, running against real market data but with virtual capital. Goal: prove profitability before going live.

### 1.1 System Coordinates

Table 1.1: Runtime coordinates

| Component   | Technology                           | Port / URL                      |
|-------------|--------------------------------------|---------------------------------|
| Frontend    | React 18 + Vite + Zustand            | 5173 (dev)                      |
| Backend     | Python 3.12 / Flask                  | 5001                            |
| WSGI (dev)  | Waitress                             | <code>serve_windows.py</code>   |
| WSGI (prod) | Gunicorn + Nginx                     | <code>gunicorn.config.py</code> |
| Database    | PostgreSQL / Alembic                 | local / RDS                     |
| Deployment  | AWS Lightsail / Ubuntu               | <code>leveragequant.tech</code> |
| Service     | <code>systemd options-trading</code> | <code>/opt/options</code>       |

### 1.2 Deployment Status (as of 2026-08-12)

- Live at [leveragequant.tech](https://leveragequant.tech)
- 26 registered users in `users_auth`
- Current HEAD: commit `12eb10b` (session 34 — expiry validation gate + staleness hard block)

- Pending deploy of session-34 fixes from dev machine
- 29 invalid paper orders (IDs 151–179) awaiting DB correction

# Chapter 2

## System Architecture

### 2.1 High-Level Block Diagram

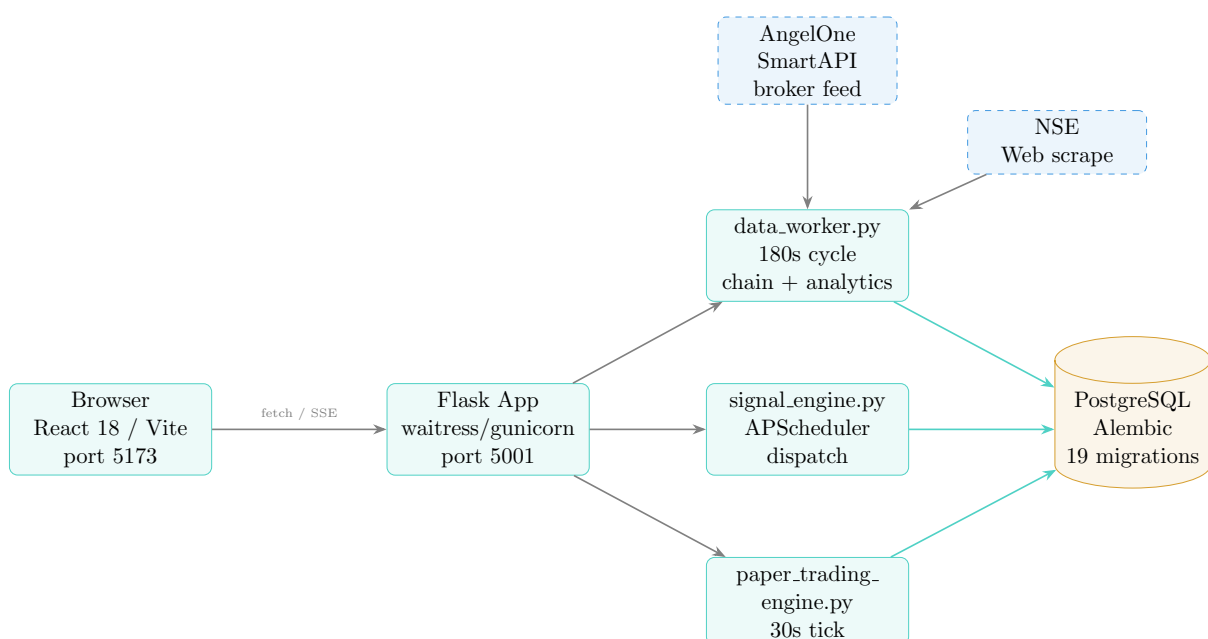


Figure 2.1: High-level system architecture

### 2.2 Request Flow

HTTP requests from the React frontend arrive at Flask. Blueprint registration in `app_factory.py` routes them to the appropriate handler. The background scheduler (APScheduler, in-process — no Celery/Redis) runs two major jobs:

1. **Signal cycle** (~180s): `data_worker.py` fetches the option chain via AngelOne's `get_option_greeks()` API, computes all analytics (GEX, DEX, VEX, Charm, EIFE, flow, OI microstructure, chain analytics), caches results in `greeks_cache` and persists snapshots to history tables.
2. **Paper trading monitor** (30s): `paper_trading_engine.py` checks open orders for SL/TP/trailing/exhaustion exits and dispatches new signals from the signal engine to qualifying users.

## 2.3 Authentication Architecture

- Stateless JWT tokens (access + refresh)
- Shared Postgres token blacklist (`token_blacklist` table)
- Rate limiting on login/refresh endpoints
- Decorator `@auth.require_auth` guards every paper-trading and analytics route
- Beta access system: `beta_access_codes` + `user_beta_access` tables (migration 019)

# Chapter 3

## Signal Generation Architecture

The signal engine is the analytical core of the platform. Its output drives both the analytics dashboard and the paper trading engine.

### 3.1 Five-Pillar Scoring Pipeline

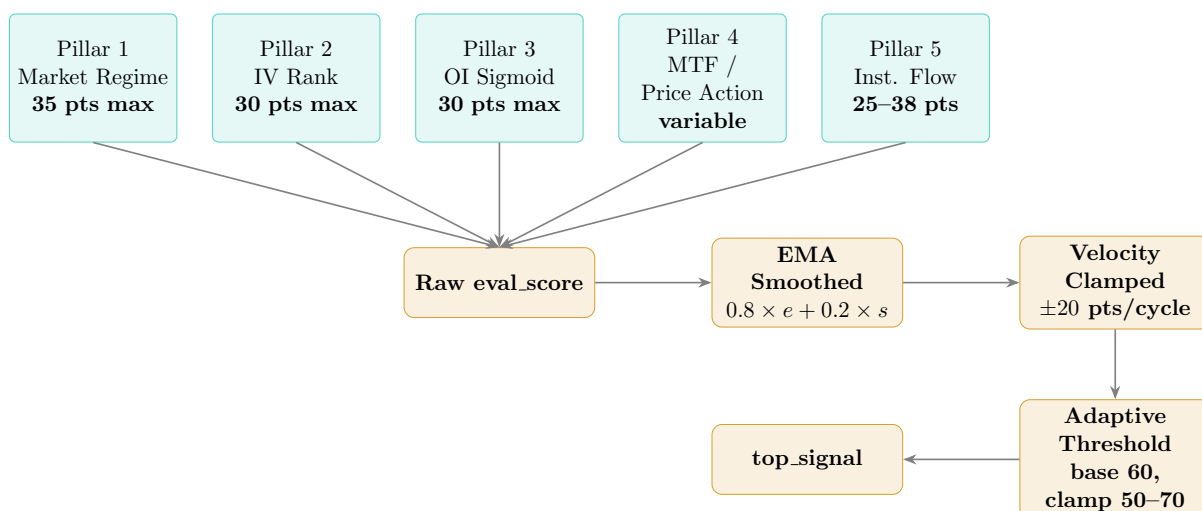


Figure 3.1: Five-pillar scoring pipeline (source: `signal_generator.py`)

#### 3.1.1 Pillar 1 — Market Regime (35 pts max)

Source: `src/strategies/market_regime.py`, `signal_generator.py`

ADX-based regime classification with hysteresis. Regime drives both the pillar score and pillar 5 (institutional flow) weighting.

Table 3.1: Market Regime ADX thresholds

| Regime Label | ADX Threshold                    | Description       | Score Contribution |
|--------------|----------------------------------|-------------------|--------------------|
| VERY_SHARP   | $\geq 45$                        | Strongly trending | Max regime score   |
| SHARP        | $\geq 30$                        | Trending          | High score         |
| MODERATE     | $\geq 22$                        | Moderate trend    | Medium score       |
| WEAK         | $\geq 18$                        | Weak trend        | Low score          |
| Choppy       | direction flips in 30-min window | Ranging           | Reduced            |

**Hysteresis parameters** (from `market_regime.py`):

- Up-threshold: 24, Down-threshold: 20
- Hysteresis margin: 2.0 pts
- Requires 2-consecutive-cycle confirmation before regime changes

**ADX computation periods:**

- 15-minute: `period=7`
- 5-minute: `period=5`
- Default: `period=14`

### 3.1.2 Pillar 2 — IV Rank (30 pts max)

Source: `src/strategies/iv_rank.py`

`IVRankService(period_days=252)`

The IV Rank formula:

$$\text{IV Rank} = \frac{IV_{\text{current}} - IV_{\text{min}}}{IV_{\text{max}} - IV_{\text{min}}} \times 100 \quad (3.1)$$

where  $IV_{\text{min}}$  and  $IV_{\text{max}}$  are computed over the trailing 252 trading days. Fallback value: 50.0 when insufficient history exists.

### 3.1.3 Pillar 3 — OI Sigmoid (30 pts max)

Source: `signal_generator.py`

Sigmoid transform of the call/put OI ratio.

$$\text{OI Score} = \frac{30}{1 + e^{-8 \cdot (r - 0.85)}} \quad (3.2)$$

where  $r$  is the call/put OI ratio. The sigmoid is centred at  $r = 0.85$  with steepness parameter  $k = 8$ .

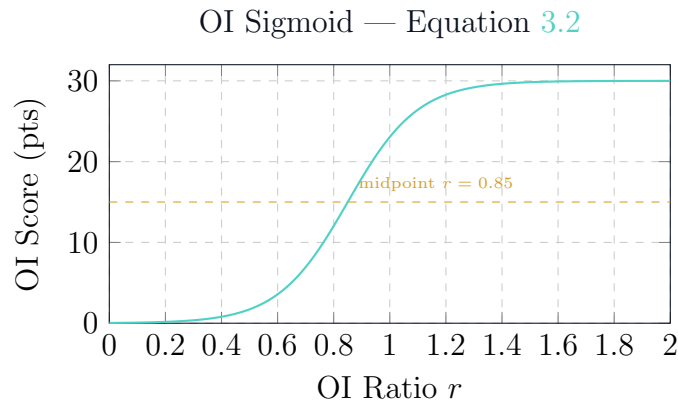


Figure 3.2: OI Sigmoid scoring curve

### 3.1.4 Pillar 4 — Multi-Timeframe / Price Action

Source: `src/strategies/analyzers/mtf_aligner.py`, `src/strategies/analyzers/price\_action.py`

Multi-timeframe alignment score. Variable contribution depending on regime.

### 3.1.5 Pillar 5 — Institutional Flow (25–38 pts max)

Source: `src/strategies/institutional_flow.py`

7 flow components, maximum contribution 38 pts. Weight influenced by market regime.

## 3.2 EMA Smoothing and Velocity Clamp

Per `signal_generator.py`, each `(strike, option_type)` key maintains a smoothed score:

$$s_t = 0.80 \cdot e_t + 0.20 \cdot s_{t-1} \quad (3.3)$$

where  $e_t$  is the raw `eval_score` at cycle  $t$  and  $s_{t-1}$  is the previous smoothed value. A velocity clamp of  $\pm 20$  pts/cycle is applied after smoothing to prevent score whipsawing.

## 3.3 Adaptive Threshold

Source: `src/strategies/generators/base_scorer.py`, `function calculate_adaptive_threshold()`

$$T = \text{clamp}(T_{\text{base}} + \Delta_{\text{VIX}} + \Delta_{\text{idle}} + \Delta_{\text{time}}, 50, 70) \quad (3.4)$$

Base:  $T_{\text{base}} = 60$ . Adjustments applied for VIX level, idle-cycle count, and time-of-day proximity to market open/close.

### 3.4 Direction Debounce and Hysteresis

- 8-point buffer before direction changes
- 3-cycle confirmation required for a direction flip
- Direction confidence gap: 10 pts minimum between leading and trailing candidates
- Cross-symbol contamination fix (session 9): all 18 hysteresis attributes stored in `_per_symbol_state[symbol]` dict, not flat instance variables

### 3.5 Signal Taxonomy

Table 3.2: Canonical signal types post session-11 normalization

| Signal    | Category    | Execution                           |
|-----------|-------------|-------------------------------------|
| BUY_CE    | DIRECTIONAL | Long call option                    |
| BUY_PE    | DIRECTIONAL | Long put option                     |
| THETA_CE  | THETA       | Short call option (premium selling) |
| THETA_PE  | THETA       | Short put option (premium selling)  |
| NO_SIGNAL | —           | No trade                            |
| HOLD      | —           | Hold existing position              |

#### Note: renamed signals

SELL\_CE and SELL\_PE were renamed to THETA\_CE and THETA\_PE in session 11 (2026-07-13) for taxonomic clarity. All code references updated.

# Chapter 4

## Quantitative Models

### 4.1 Black-Scholes-Merton Framework

Source: `src/core/black_scholes.py`

Full BSM Greeks implementation with Newton-Raphson IV solver.

#### 4.1.1 Option Pricing

Let  $S$  = spot price,  $K$  = strike,  $T$  = time to expiry (years),  $r$  = risk-free rate,  $q$  = dividend yield,  $\sigma$  = implied volatility.

The BSM forward price:

$$F = S \cdot e^{(r-q)T} \quad (4.1)$$

**Implementation defaults:**  $r = 6.5\%$ , 365-day calendar year.

$$d_1 = \frac{\ln(S/K) + (r - q + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}} \quad (4.2)$$

$$d_2 = d_1 - \sigma\sqrt{T} \quad (4.3)$$

Call price:

$$C = Se^{-qT}N(d_1) - Ke^{-rT}N(d_2) \quad (4.4)$$

Put price:

$$P = Ke^{-rT}N(-d_2) - Se^{-qT}N(-d_1) \quad (4.5)$$

### 4.1.2 Greeks

$$\Delta_{\text{call}} = e^{-qT} N(d_1) \quad (4.6)$$

$$\Delta_{\text{put}} = -e^{-qT} N(-d_1) \quad (4.7)$$

$$\Gamma = \frac{e^{-qT} N'(d_1)}{S\sigma\sqrt{T}} \quad (4.8)$$

$$\Theta_{\text{call}} = -\frac{S\sigma e^{-qT} N'(d_1)}{2\sqrt{T}} - rKe^{-rT} N(d_2) + qSe^{-qT} N(d_1) \quad (4.9)$$

$$\mathcal{V} = Se^{-qT} N'(d_1)\sqrt{T} \quad (4.10)$$

$$\rho_{\text{call}} = KTe^{-rT} N(d_2) \quad (4.11)$$

### 4.1.3 Newton-Raphson IV Solver

#### Found in Repository

IV solver iterates until convergence. IV is clamped to range [1%, 500%].

$$\sigma_{n+1} = \sigma_n - \frac{f(\sigma_n)}{f'(\sigma_n)} = \sigma_n - \frac{C_{\text{BSM}}(\sigma_n) - C_{\text{market}}}{\mathcal{V}(\sigma_n)} \quad (4.12)$$

## 4.2 Gamma Exposure (GEX)

Source: `scripts/engine/gex_engine.py`

GEX = per-strike net gamma exposure in rupee terms.

### 4.2.1 Lot Sizes

Analytics `LOT_SIZES` (from `gex_engine.py`):

| Instrument | Lot Size |
|------------|----------|
| NIFTY      | 75       |
| BANKNIFTY  | 40       |
| FINNIFTY   | 75       |

Paper Trading `LOT_SIZES` (from `paper_trading_engine.py`):

| Instrument | Lot Size |
|------------|----------|
| NIFTY      | 65       |
| BANKNIFTY  | 30       |
| FINNIFTY   | 40       |
| MIDCPNIFTY | 75       |

### 4.2.2 GEX Formula

$$\text{GEX}_{\text{strike}} = |\Gamma_{\text{strike}}| \times OI_{\text{strike}} \times \text{lot\_size} \quad (4.13)$$

$$\text{Net GEX} = \text{GEX}_{\text{puts}} - \text{GEX}_{\text{calls}} \quad (4.14)$$

- **Gamma Regime Threshold:**  $50,000,000 \times (1 - \text{dte}/30)$ ;  $\text{DTE} > 30 \Rightarrow \text{NEUTRAL}$
- **Pin Strike:**  $\text{DTE} \leq 2$ , within 2% of spot, highest  $\Gamma \times OI$  concentration. Returns None if conditions not met.

## 4.3 EIFE — Enhanced Institutional Flow Engine

Source: `scripts/engine/eife_engine.py`

EIFE aggregates multiple flow signals into a single directional score.

### 4.3.1 IFS Formula

$$\text{IFS} = \text{clamp}\left(\frac{\text{net\_strength}}{\text{MAX\_DIRECTIONAL\_PILLARS}} \times 10, -10, +10\right) \quad (4.15)$$

- BULLISH if  $\text{IFS} > 2$
- BEARISH if  $\text{IFS} < -2$
- NEUTRAL otherwise

### 4.3.2 Confidence Modifiers

$$\text{CONF\_RVOL\_MAX\_PCT} = 0.10 \quad (4.16)$$

$$\text{CONF\_HHI\_MAX\_PCT} = 0.10 \quad (4.17)$$

$$\text{CONF\_ND\_MAX\_PCT} = 0.10 \quad (4.18)$$

The `net_delta_oi_zscore` (session 6 fix) now modulates confidence as a corroborating signal: confidence rises when its sign agrees with the overall vote direction.

## 4.4 Flow Engine

Source: `scripts/engine/flow_engine.py`

**PCR (Put/Call Ratio):**

$$\text{PCR} = \frac{\sum OI_{\text{puts}}}{\sum OI_{\text{calls}}} \quad (4.19)$$

**HHI (Herfindahl-Hirschman Index) — OI concentration:**

$$\text{HHI} = \sum_i \left( \frac{OI_i}{\sum_j OI_j} \right)^2 \quad (4.20)$$

## 4.5 Theta Scorer

Source: `src/strategies/theta_scorer.py`

Drives THETA signal dispatch. OI wall gate was **removed** 2026-07-15; only VIX, ADX, and IV\_rank now contribute.

Table 4.1: VIX score contribution

| VIX Condition | Score (pts) |
|---------------|-------------|
| $\geq 18$     | 25          |
| $\geq 14$     | 15          |
| $\geq 11$     | 8           |
| $< 11$        | 0           |

Table 4.2: ADX score contribution (theta)

| ADX Condition | Score (pts) |
|---------------|-------------|
| $< 18$        | 15          |
| $< 22$        | 10          |
| $< 28$        | 5           |

**Dispatch threshold:** `theta_score`  $\geq 20$  (lowered from 40 in session 29, 2026-07-28).

# Chapter 5

## SL/TP Selection Engine

```
Source:  src/strategies/analyzers/trade_orchestrator.py,  scripts/server/
paper\_trading\_engine.py
```

### 5.1 Hard Floor (session 13, 2026-07-13)

Every SL candidate must clear a hard minimum distance floor. The floor is the maximum of three components:

$$\text{floor} = \max( d_{\text{flat}}, d_{\text{pct}}, d_{\text{ATR}} ) \quad (5.1)$$

where:

$$d_{\text{flat}} = 10.0 \text{ pts} \quad (\text{\_MIN\_DISTANCE\_FLOOR\_PTS}) \quad (5.2)$$

$$d_{\text{pct}} = 0.12 \times \text{entry\_premium} \quad (\text{\_MIN\_DISTANCE\_FLOOR\_PCT}) \quad (5.3)$$

$$d_{\text{ATR}} = \text{ATR} \times 0.5 \times |\Delta| \quad (5.4)$$

This floor is applied as a **hard exclusion** (not a score penalty) in `_select_best_sl`, `_select_targets`, `_weak_confluence_fallback`, and `_pct_fallback_tp_sl` (session 14 fix — the config-% fallback path).

### 5.2 SL/TP Source Hierarchy

Table 5.1: SL/TP candidate sources with weights

| Source                   | Type        | Weight                  |
|--------------------------|-------------|-------------------------|
| SMC FVG (Fair Value Gap) | SL + TP     | 10                      |
| SMC Order Block          | SL + TP     | 8 (quality-scaled)      |
| Liquidity Sweep          | SL          | 8                       |
| Gamma Flip Level         | TP          | 7                       |
| OI Wall (call/put)       | TP          | $\propto$ walls_quality |
| ATR Extension            | Fallback    | n/a                     |
| Weak Confluence          | Last-resort | n/a                     |

**Confluence bonus:** +15 pts per source within 0.15% of another.

**TP cap:** 40% premium gain maximum.

**Gate 6:** R:R is recomputed from the *final* `tp1/sl` at fill time (not the pre-override estimate). Cap at 10:1.

## 5.3 Trailing SL Constants

$$\text{\_MIN\_TRAILING\_DISTANCE\_PCT} = 12.0\% \quad (5.5)$$

$$\text{\_TP\_CAPTURE\_PCT} = 0.98 \quad (\text{within } 2\% \text{ of TP counts as hit}) \quad (5.6)$$

The trailing SL formula (both slow path and WS fast path):

$$\text{new\_sl} = \text{lt}p - \frac{\text{fill} \times d\%}{100} \quad (5.7)$$

floored at `fill_price` (breakeven floor invariant, session 6 fix).

## Chapter 6

# Paper Trading Safety Gate System

Source: `scripts/server/paper_trading_engine.py`

All gates live in `_safety_check()` except where noted. Executed sequentially; first failure blocks the trade.

## 6.1 Gate Waterfall Diagram

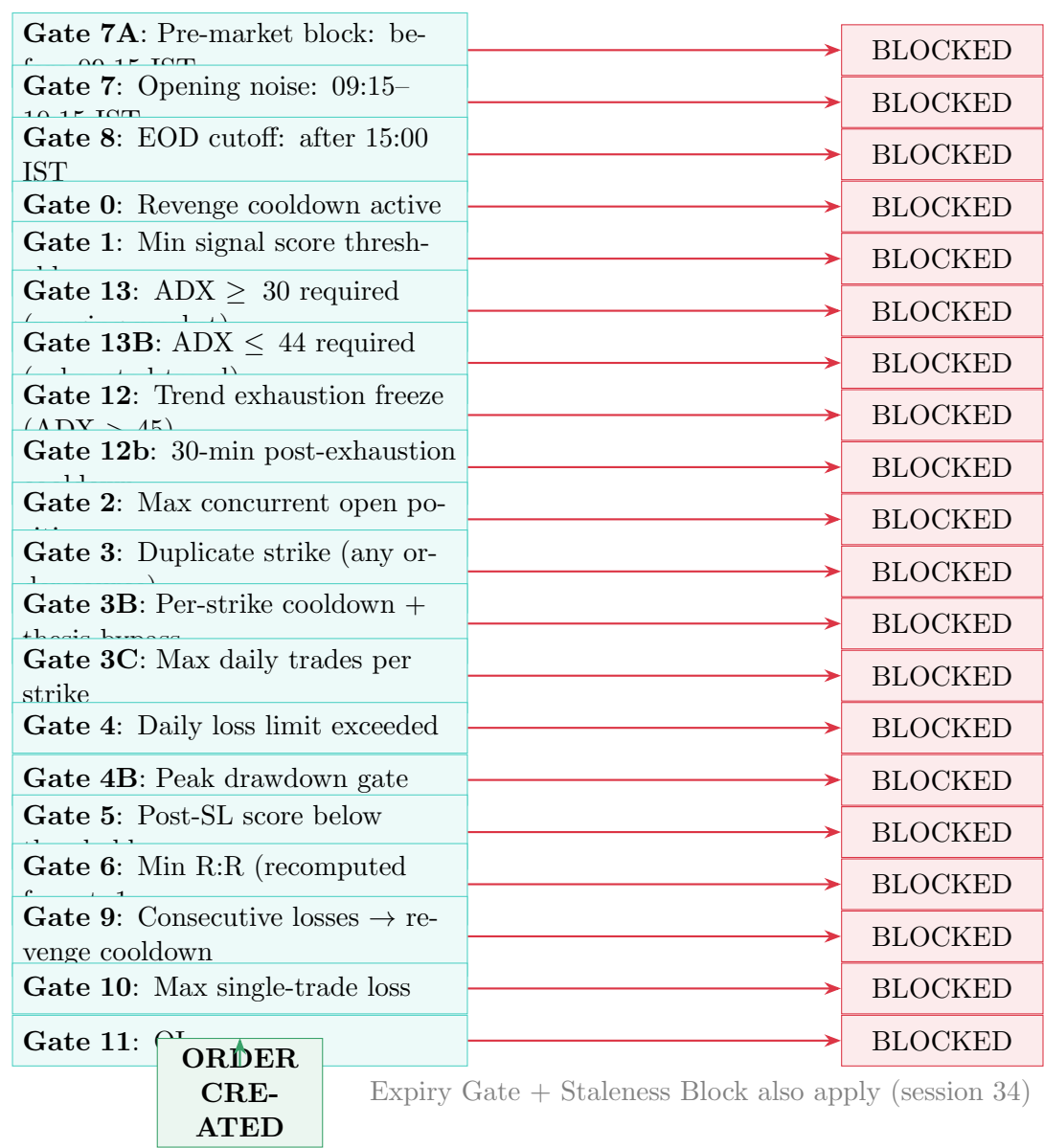


Figure 6.1: Safety gate waterfall — all 20 gates (source: `_safety_check()`)

sectionGate Details

Table 6.1: Paper trading safety gates — key parameters

| Gate                         | Condition                                                             | Notes                                        |
|------------------------------|-----------------------------------------------------------------------|----------------------------------------------|
| 7A                           | <code>now_time &lt; 09:15 IST</code>                                  | Pre-market block; added session 28           |
| 7                            | <code>09:15--10:15 IST</code>                                         | Opening noise block; extended in session 28  |
| 8                            | <code>now_time &gt; 15:00 IST</code>                                  | EOD cutoff                                   |
| 0                            | <code>revenge\_cooldown\_until</code> active                          | Set by Gate 9; visible in ExecStatus         |
| 1                            | <code>score &lt; min_signal_score</code>                              | Per-user config; also expiry-day min score   |
| 13                           | <code>ADX &lt; 30</code>                                              | Ranging market; DIRECTIONAL only             |
| 13B                          | <code>ADX &gt; 44</code>                                              | Exhausted trend (was 45, lowered session 28) |
| 12                           | <code>TREND\_EXHAUSTION\_EXIT</code> today + <code>ADX &gt; 45</code> | Same (symbol, direction) blocked             |
| 12b                          | Within 30 min of <code>TREND\_EXHAUSTION\_EXIT</code>                 | Even after ADX drops below 45                |
| 2                            | <code>Open orders ≥ max\_concurrent\_orders</code>                    | Also checked at pending-fill time            |
| 3                            | Open order at same strike+type exists                                 | Any order_source (ALGO or MANUAL)            |
| 3B                           | Within cooldown window + OI verdict                                   | Thesis-aware bypass: 3 re-entries max        |
| 3C                           | Daily ALGO trades at strike ≥ limit                                   | Per-strike daily cap                         |
| 4                            | Today PnL loss ≥ <code>max\_daily\_loss\_pct</code>                   | Uses start-of-day balance as denominator     |
| 4B                           | $(peak - equity)/peak ≥ max\_drawdown\_pct$                           | Equity = cash + unrealised PnL               |
| 5                            | <code>Score &lt; post\_sl\_score\_threshold</code>                    | After a SL hit on same symbol                |
| 6                            | Recomputed R:R < <code>min_rr_ratio</code>                            | Cap at 10:1                                  |
| 9                            | Consecutive losses ≥ limit                                            | Sets revenge cooldown on trigger             |
| 10                           | Potential loss > <code>max\_single\_trade\_loss\_pct</code>           | Uses <code>abs()</code> for long/short       |
| 11                           | OI/Flow microstructure OPPOSE verdict                                 | <code>_get_oi_gate_verdict()</code>          |
| <i>Session 34 additions:</i> |                                                                       |                                              |
| Expiry                       | Parsed expiry date < today                                            | Blocks expired contracts                     |
| Stale                        | <code>chain_age_secs &gt; 1800</code>                                 | Hard blocks new fills on stale chain         |

## 6.2 Thesis-Aware Re-Entry (Gate 3B Bypass)

Source: `paper_trading_engine.py`, constants near line 117

- `_THESIS_INTACT_EXITS`: {`TP_HIT`, `BREAKEVEN_HIT`, `TRAILING_SL_HIT`, `HOLD_EXIT`, `TIME_EXIT`, `PRICE_REVERSION`}
- `_THESIS_REENTRY_CAP` = 3 (1 original + 2 re-entries)
- Bypass only if: last exit is thesis-intact  $\wedge$  daily ALGO count at strike  $< 3 \wedge$  OI verdict is not OPPOSE

## 6.3 Trend Exhaustion Exit (Gate 12)

Source: `\_maybe\_exit\_on\_trend\_exhaustion()`, `src/strategies/generators/base\_scorer.py::calculate\_exhaustion()`

- Fires when `exhaustion.status == "EXHAUSTION"` (`ADX > 45`)
- **Unconditional**: any P&L, confirmed live 2026-07-14
- Per-user toggle: `exhaustion_exit_enabled` (migration 018, default `TRUE`)
- Exit reason: `TREND_EXHAUSTION_EXIT`
- Gate 12b: additional 30-minute cooldown after `ADX` drops below 45; constant `_EXHAUSTION_COOLDOWN_MINS = 30`

## 6.4 BLIND\_CLOSE — Missing LTP Escalation

Source: `paper_trading_engine.py`

Table 6.2: `BLIND_CLOSE` escalation tiers

| Ticks | Constant                       | Action                                                                        |
|-------|--------------------------------|-------------------------------------------------------------------------------|
| 1–4   | <code>_BLIND_WARN = 5</code>   | WARNING log                                                                   |
| 5–19  |                                | ERROR log                                                                     |
| 20+   | <code>_BLIND_CLOSE = 20</code> | Force-close at <code>fill_price</code> ; exit reason <code>BLIND_CLOSE</code> |

# Chapter 7

## Exit Reasons and Position Lifecycle

Source: `paper_trading_engine.py`, `EXIT_COLORS` in `PaperTradingHistory.jsx`

### 7.1 Enumeration of Exit Reasons

Table 7.1: All 15 paper trading exit reasons

| Exit Reason             | Path                                                | Notes                                        |
|-------------------------|-----------------------------------------------------|----------------------------------------------|
| SL_HIT                  | Monitor / WS fast path                              | Closed at stated SL price (not live LTP)     |
| TP_HIT                  | Monitor / WS fast path                              | Closed at <code>min(ltp, tp)</code> for long |
| BREAKEVEN_HIT           | Monitor                                             | SL was at or above fill price                |
| TRAILING_SL_HIT         | Monitor / WS                                        | Trail SL floored at fill                     |
| HOLD_EXIT               | Monitor                                             | HOLD_PROFIT_EXIT; after hold window          |
| TIME_EXIT               | Monitor                                             | EOD square-off                               |
| MANUAL_CLOSE            | API route                                           | User-initiated                               |
| PRICE_REVERSION         | Monitor                                             | Entry thesis reversed                        |
| SIGNAL_FLIP             | <code>\_handle\_signal\_flip()</code>               | Opposing direction signal                    |
| TREND_EXHAUSTION_EXIT   | <code>\_maybe\_exit\_on\_trend\_exhaustion()</code> | ADX > 45                                     |
| CONSECUTIVE_LOSSES_EXIT | Gate 9 trigger                                      | Account protection                           |
| MAX_LOSS_EXIT           | Per-trade monitor (step 3.6)                        | Unrealised loss exceeds limit                |
| DRAWDOWN_EXIT           | Gate 4B trigger                                     | Peak drawdown exceeded                       |
| REVENGE_COOLDOWN_EXIT   | Gate 0 trigger                                      | Cooldown window active                       |
| BLIND_CLOSE             | Missing LTP monitor                                 | 20 consecutive None-LTP ticks                |

## 7.2 Signal Flip Conflict Matrix

Source: `_FLIP_PAIRS`, `_conflicts_with_new_signal()`, session 17

Table 7.2: Signal flip pairs — 6 total pairs

| Open Position | New Signal | Action                                |
|---------------|------------|---------------------------------------|
| BUY_CE        | BUY_PE     | FLIP (close CE, enter PE)             |
| BUY_PE        | BUY_CE     | FLIP                                  |
| THETA_CE      | BUY_CE     | FLIP (same side, opposing strategy)   |
| BUY_CE        | THETA_CE   | FLIP                                  |
| THETA_PE      | BUY_PE     | FLIP                                  |
| BUY_PE        | THETA_PE   | FLIP                                  |
| THETA_CE      | THETA_PE   | <b>NO FLIP</b> (both premium-selling) |
| BUY_CE        | THETA_PE   | <b>NO FLIP</b> (cross-type same lean) |

# Chapter 8

## Database Architecture

### 8.1 Migration Timeline

Table 8.1: Alembic migration history (19 total)

| Migration | Focus           | Key Tables/Changes                                                                                                                                |
|-----------|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| 001–003   | Base + Auth     | Market data tables, auth tables ( <code>users_auth</code> , <code>token_blacklist</code> , <code>login_attempts</code> , <code>audit_log</code> ) |
| 004       | Paper accounts  | <code>paper_trading_accounts</code>                                                                                                               |
| 005       | Paper orders    | <code>paper_auto_config</code> , <code>paper_orders</code> , <code>paper_order_events</code>                                                      |
| 006–010   | Gate extensions | Trailing SL, breakeven SL, RR gate, HOLD, consecutive-loss gates, drawdown, revenge cooldown, peak-balance                                        |
| 011       | Cleanup         | Drop <code>paper_positions</code> , <code>paper_trade_history</code> (superseded by <code>paper_orders</code> )                                   |
| 012–014   | Risk gates      | Smart re-entry, daily reset, expiry-day risk, perf indexes                                                                                        |
| 015       | Auth            | Refresh-token lookup                                                                                                                              |
| 016       | Paper hold      | <code>paper_hold_state</code> (persist <code>hold_start</code> )                                                                                  |
| 017       | Constraint      | <code>chk_virtual_balance_nonneg</code> on <code>paper_trading_accounts</code>                                                                    |
| 018       | Exhaustion      | <code>exhaustion_exit_enabled</code> column on <code>paper_auto_config</code>                                                                     |
| 019       | Beta access     | <code>beta_access_codes</code> , <code>user_beta_access</code>                                                                                    |

### 8.2 Core Paper Trading Tables

### 8.2.1 paper\_trading\_accounts

Table 8.2: paper\_trading\_accounts key columns

| Column                   | Type         | Notes                              |
|--------------------------|--------------|------------------------------------|
| user_id                  | UUID         | FK → users_auth                    |
| virtual_balance          | NUMERIC      | CHECK ≥ 0 (migration 017)          |
| peak_balance             | NUMERIC      | All-time high; Gate 4B denominator |
| revenge\_cooldown\_until | TIMESTAMP TZ | Gate 0 expiry                      |

Initial balance: ₹1,000,000 (INITIAL\_BALANCE)

### 8.2.2 paper\_orders (single source of truth)

Table 8.3: paper\_orders key columns

| Column          | Type         | Notes                                      |
|-----------------|--------------|--------------------------------------------|
| id              | SERIAL       | Primary key                                |
| user_id         | UUID         | FK → users_auth                            |
| symbol          | TEXT         | NIFTY / BANKNIFTY / FINNIFTY               |
| strikeprice     | NUMERIC      | Integer strike                             |
| optiontype      | TEXT         | CE / PE                                    |
| transactiontype | TEXT         | BUY (long) / SELL (short/theta)            |
| lots            | INT          |                                            |
| lotsize         | INT          | Snapshot at fill time                      |
| fill_price      | NUMERIC      | Entry premium                              |
| stoploss        | NUMERIC      | Current SL                                 |
| target_price    | NUMERIC      | Current TP                                 |
| exit_price      | NUMERIC      | Exit premium (capped at target for TP_HIT) |
| pnl             | NUMERIC      | Realised P&L                               |
| status          | TEXT         | open / pending / complete                  |
| order_source    | TEXT         | ALGO / MANUAL                              |
| exit_reason     | TEXT         | One of 15 exit reasons                     |
| signal_id       | VARCHAR(200) | Dispatch signal identifier                 |
| trailing_active | BOOLEAN      | Trail/breakeven engaged                    |

paper\_positions and paper\_trade\_history (created migration 004) were **super-seded** by paper\_orders in migration 005 and **formally dropped** in migration 011. Do not resurrect these tables.

## 8.3 Analytics History Tables

Table 8.4: Analytics history tables

| Table                            | Contents                                                   |
|----------------------------------|------------------------------------------------------------|
| <code>gex.history</code>         | Per-cycle GEX, DEX, VEX, Charm snapshots                   |
| <code>flow.history</code>        | PCR, HHI, walls, micro-ATM snapshots                       |
| <code>eife.history</code>        | IFS, signal, confidence per symbol per cycle               |
| <code>vol.surface.history</code> | IV surface snapshots (up to 30-min stale)                  |
| <code>vex.dex.history</code>     | DEX/VEX/Charm aggregates including net Theta/Vega          |
| <code>greeks.cache</code>        | Live chain from AngelOne ( <code>greeks.chain</code> JSON) |
| <code>signal.archive</code>      | Every signal cycle record                                  |
| <code>strike.oi.history</code>   | Per-strike OI over time (multi-straddle)                   |
| <code>intraday.candles.5m</code> | 5-minute OHLCV candles (ticker.id=1 NIFTY, 2 BANKNIFTY)    |

# Chapter 9

## Alpha Discovery

### 9.1 Alpha Discovery 1.0 — Structural Profitability Audit

Source: CLAUDE.md §8 sessions 27–28 (2026-07-24)

All findings verified against live production DB before implementation.

#### 9.1.1 Baseline (All-Time, Pre-Fix)

- Total ALGO trades: 96
- All-time P&L: **−₹4,72,736**
- SL\_HIT: 21 trades, TARGET\_HIT: 3 trades, BREAK EVEN\_HIT: 18 trades

#### 9.1.2 Root Causes Identified

Table 9.1: Five root causes of profitability failure

| Finding                                  | Evidence                                    | Fix                                       |
|------------------------------------------|---------------------------------------------|-------------------------------------------|
| SL exit used live LTP, not stated SL     | 10s monitoring gap; LTP can be ₹80 below SL | Close at stated SL price (commit a0d2537) |
| Entries during 09:15–10:15 opening noise | 5 BREAK EVEN in first 20 min, 0% win rate   | Gate 7 extended to 10:15 (commit 188d7d9) |
| Post-exhaustion re-entries < 30 min      | 27 trades avg −₹13,278 vs −₹1,655 for clean | Gate 12b 30-min cooldown (commit 312fbdf) |
| ADX < 30 entries (ranging market)        | 7/10 SL_HITs had ADX < 30                   | Gate 13 (commit cb9a44c)                  |
| ADX > 45 entries (exhausted trend)       | Dead-on-arrival in 3–5s                     | Gate 13B at 44 (commits d44d133, f4054c2) |

### 9.1.3 Full Retroactive Simulation (All Fixes Applied)

Table 9.2: Simulated P&L with Gate 13B = 44, all fixes applied

| Exit Reason            | Trades    | Corrected P&L (₹)         |
|------------------------|-----------|---------------------------|
| BREAKEVEN_HIT          | 13        | +1,09,801                 |
| EXHAUSTION_OI_MICRO    | 13        | +69,794                   |
| TARGET_HIT             | 3         | +40,528                   |
| MANUAL_CLOSE           | 2         | +11,359                   |
| TREND_EXHAUSTION_EXIT  | 4         | +5,025                    |
| SIGNAL_FLIP            | 4         | −4,641                    |
| SL_HIT (3 irreducible) | 3         | −51,026                   |
| <b>TOTAL</b>           | <b>42</b> | <b>+1,80,840 (+18.1%)</b> |

Table 9.3: Performance metrics post all-fixes simulation

| Metric                                   | Value                  |
|------------------------------------------|------------------------|
| Return                                   | +18.1% on ₹10L capital |
| Profit Factor                            | 2.41                   |
| Win Rate                                 | 80.5%                  |
| Pre-live criterion (win rate > 40%)      | ✓PASS                  |
| Pre-live criterion (profit factor > 1.2) | ✓PASS                  |

### 9.1.4 Irreducible Losses

Three SL\_HIT trades remain after all gate fixes, representing genuine wrong-direction calls (cannot be prevented by structural gates):

- Order 49: BUY\_PE, ADX 33.48, score 88.3, −₹38,064 (77% of remaining)
- Orders 103/104: BUY\_CE, ADX 31–32, 14:12 IST, −₹12,858 combined

These motivate Alpha Discovery 2.0 (§9.2).

## 9.2 Alpha Discovery 2.0 — Signal Intelligence Research

Source: CLAUDE.md §11

Directive: treat signal generation as a quantitative research problem. Do not assume the signal direction is correct.

### 9.2.1 Signal Replay Engine Findings

Table 9.4: Signal replay of 5 “WOULD STILL FIRE” orders post all-fixes

| Order(s)            | Context           | Finding                                                                                                           |
|---------------------|-------------------|-------------------------------------------------------------------------------------------------------------------|
| 49 (13:15 IST)      | BUY_PE, ADX 33.48 | NIFTY flat at 23980; T+15 rallied +34pts. Genuine directional miss.                                               |
| 87/88 (11:02 IST)   | BUY_PE, ADX 44.37 | Downtrend completed <i>inside</i> Gate 7 window. Alpha half-life: 10–12 minutes. Late entry into completed trend. |
| 103/104 (14:11 IST) | BUY_CE, ADX 31–32 | NIFTY already declining from 23817; fell 33pts in 20 min. Immediate wrong-direction call.                         |

### 9.2.2 Time-of-Day Win Rate Analysis

Table 9.5: Win rate and P&L by hour IST (all historical ALGO trades)

| Hour IST | Trades | Win%       | Total P&L (₹) |
|----------|--------|------------|---------------|
| 9:xx     | 28     | 50%        | −1,50,000     |
| 10:xx    | 21     | 33%        | −12,800       |
| 11:xx    | 14     | 21%        | −4,41,000     |
| 12:xx    | 25     | <b>92%</b> | +1,68,900     |
| 13:xx    | 9      | 67%        | −700          |
| 14:xx    | 6      | 33%        | −25,900       |

The 12:xx IST window shows a dramatically superior win rate (92%) compared to all other hours. The 11:xx loss is dominated by orders 87/88 (ADX 44.37, slippage victims).

### 9.2.3 ADX Bucket Analysis

Table 9.6: ADX bucket performance — empirical basis for Gate 13B = 44

| ADX Bucket         | Trades | Win% | Total P&L (₹)    |
|--------------------|--------|------|------------------|
| 40                 | 5      | 100% | +69,924          |
| 41                 | 2      | 100% | +23,530          |
| 42                 | 6      | 83%  | +12,123          |
| 44                 | 4      | 50%  | <b>−3,14,915</b> |
| 45+ (old Gate 13B) | 2      | 0%   | −741             |

ADX 44 is catastrophically bad (avg −₹78,729/trade, dominated by orders 87/88). Gate 13B was lowered from 45 to 44 on this empirical basis.

# Chapter 10

## Session 34 — Expired Contract Incident

**Critical Incident: 2026-08-05 to 2026-08-10**

5 trading days of invalid orders due to a silent API failure.

### 10.1 Root Cause

AngelOne's `get_option_greeks()` API returned:

```
1 # HTTP 200 -- but this response body:
2 {"status": None, "message": "Invalid Token"}
```

This bypasses exception handling entirely. The engine's response-validation check if `not data or not data.get("data")` was ineffective because this response has a non-`None` `data` field. Result: `sc.greeks_chain` was preserved with **04-AUG-2026 (expired)** contracts. Every dispatch filled at the frozen stale LTP of an expired option (₹39–73), which was immediately worth  $\approx$ ₹0 at the next monitoring tick  $\Rightarrow$  SL\_HIT in 8–11 seconds, every trade, for 5 days.

### 10.2 Quantified Damage

Table 10.1: Invalid orders by user (IDs 151–179)

| User (truncated) | Orders    | Loss (₹)            |
|------------------|-----------|---------------------|
| 53caae77         | 14        | −4,02,184.90        |
| 25f56e2e         | 4         | −1,17,000.00        |
| fe6caf33         | 8         | −73,963.50          |
| 3dabb35b         | 3         | −20,256.60          |
| <b>Total</b>     | <b>29</b> | <b>−6,13,405.00</b> |

## 10.3 Fixes Deployed (Commit 12eb10b)

### 10.3.1 Fix 1 — Expiry Validation Gate

Applied in `on_signal_received()`. After `_get_nearest_expiry()`, parse to a Python date and compare against today (IST):

```
1 _expiry_date = _expiry_str_to_date(nearest_expiry)
2 if _expiry_date is not None and _expiry_date < _today_ist:
3     logger.warning("[PAPER] EXPIRY GATE BLOCKED: ...")
4     return # no dispatch
5 # Same-day expiry is ALLOWED (< not <=)
```

### 10.3.2 Fix 2 — Staleness Hard Block

Applied in `_process_signal_for_user()`. When `chain_age_secs > 1800` (30 minutes), block new fills:

```
1 if chain_age_secs > 1800:
2     logger.error("[PAPER] STALENESS HARD BLOCK: ...")
3     return # no order creation
```

Existing open positions continue to be monitored normally (only new fills are blocked).

## 10.4 Test Coverage Added

- `TestExpiryValidationGate` (5 tests): past expiry blocked, warning logged, future expiry proceeds, same-day not blocked, unparseable expiry does not block (fail-open)
- `TestStalenessHardBlock` (5 tests): >30 min blocked, error logged, exactly-at-threshold not blocked, below threshold passes, None chain\_age passes
- Suite: **1,065 passed, 5 pre-existing failures** (baseline +10)

# Chapter 11

## Analytics Platform

### 11.1 Backend Endpoints

Table 11.1: Selected analytics endpoints (`routes_analytics.py`,  $\approx 44$  total)

| Endpoint                                                      | Description                               |
|---------------------------------------------------------------|-------------------------------------------|
| <code>GET/api/analytics/gex/&lt;symbol&gt;</code>             | Gamma exposure per strike                 |
| <code>GET/api/analytics/dex/&lt;symbol&gt;</code>             | Delta exposure per strike                 |
| <code>GET/api/analytics/vex/&lt;symbol&gt;</code>             | Vanna exposure                            |
| <code>GET/api/analytics/charm/&lt;symbol&gt;</code>           | Charm exposure                            |
| <code>GET/api/analytics/theta-vega/&lt;symbol&gt;</code>      | Net Theta/Vega aggregate (new session 16) |
| <code>GET/api/analytics/flow/&lt;symbol&gt;</code>            | PCR, HHI, walls                           |
| <code>GET/api/analytics/eife/&lt;symbol&gt;</code>            | EIFE signal and confidence                |
| <code>GET/api/analytics/chain-analytics/&lt;symbol&gt;</code> | Full option chain with Greeks             |
| <code>GET/api/analytics/vol-surface/&lt;symbol&gt;</code>     | IV surface by expiry                      |
| <code>GET/api/analytics/screener</code>                       | Filtered option chain screener            |
| <code>GET/api/analytics/strategy-wizard/&lt;symbol&gt;</code> | Strategy recommendations                  |
| <code>POST/api/analytics/strategy-builder/preview</code>      | Payoff curve + Greeks                     |
| <code>GET/api/analytics/oi-heatmap/&lt;symbol&gt;</code>      | OI by strike/expiry                       |
| <code>GET/api/analytics/expiry-calendar</code>                | NSE expiry dates                          |

## 11.2 Frontend Sections

Source: `frontend/src/features/analytics/sections/`

25+ React sections, each behind React Router lazy-loading. Maximum 2–4 concurrent analytics polls at any time (not 25+; confirmed session 7).

Table 11.2: Key frontend analytics sections

| Section Component                     | Focus                                                               |
|---------------------------------------|---------------------------------------------------------------------|
| <code>GammaExposureSection</code>     | GEX per strike, gamma flip level, pin strike                        |
| <code>DealerPositioningSection</code> | EIFE, confidence breakdown, Greeks summary                          |
| <code>FlowSection</code>              | PCR, HHI, OI walls, micro-ATM liquidity                             |
| <code>ChainAnalyticsSection</code>    | Full chain with Greeks view (Delta/Gamma/Theta/Vega), ITM shading   |
| <code>VolSurfaceSection</code>        | IV smile by expiry (OTM-preferred convention)                       |
| <code>StrategyBuilderSection</code>   | Manual leg builder, live premium auto-fill, payoff + Greeks         |
| <code>StrategyWizardSection</code>    | 20 templates ranked by POP + IV fit score                           |
| <code>ScreenersSection</code>         | Filtered chain with LTP/Delta/Theta/Vega columns                    |
| <code>MultiStraddleSection</code>     | Multi-expiry straddle P&L chart                                     |
| <code>OIHeatmapSection</code>         | Strike $\times$ expiry OI heatmap                                   |
| <code>ExpiryCalendarSection</code>    | NSE holiday and expiry calendar                                     |
| <code>TechnicalSignalsSection</code>  | Technical signal summary with <code>last_close</code> and timeframe |

## 11.3 Polling Architecture

Table 11.3: Frontend poll intervals vs backend refresh cadence

| Data Type        | Frontend Poll | Backend Refresh                     |
|------------------|---------------|-------------------------------------|
| Analytics (most) | 20s           | 180s                                |
| FII/DII          | 120s          | 3600s (1 hour)                      |
| Vol Surface      | 120s          | 1800s (30 min)                      |
| ExecStatus       | 5s            | Real-time DB query                  |
| SignalDiagnostic | 10s           | In-memory <code>_last_result</code> |

**Visibility awareness:** all polls use `pollWhileVisible.js` (session 7 fix). A catch-up poll fires on tab-focus return only if a full interval has elapsed since the last run (prevents CDP/automation spurious toggle exploitation).

## Chapter 12

# Paper Trading Engine — Operational Detail

## 12.1 Dispatch Flow

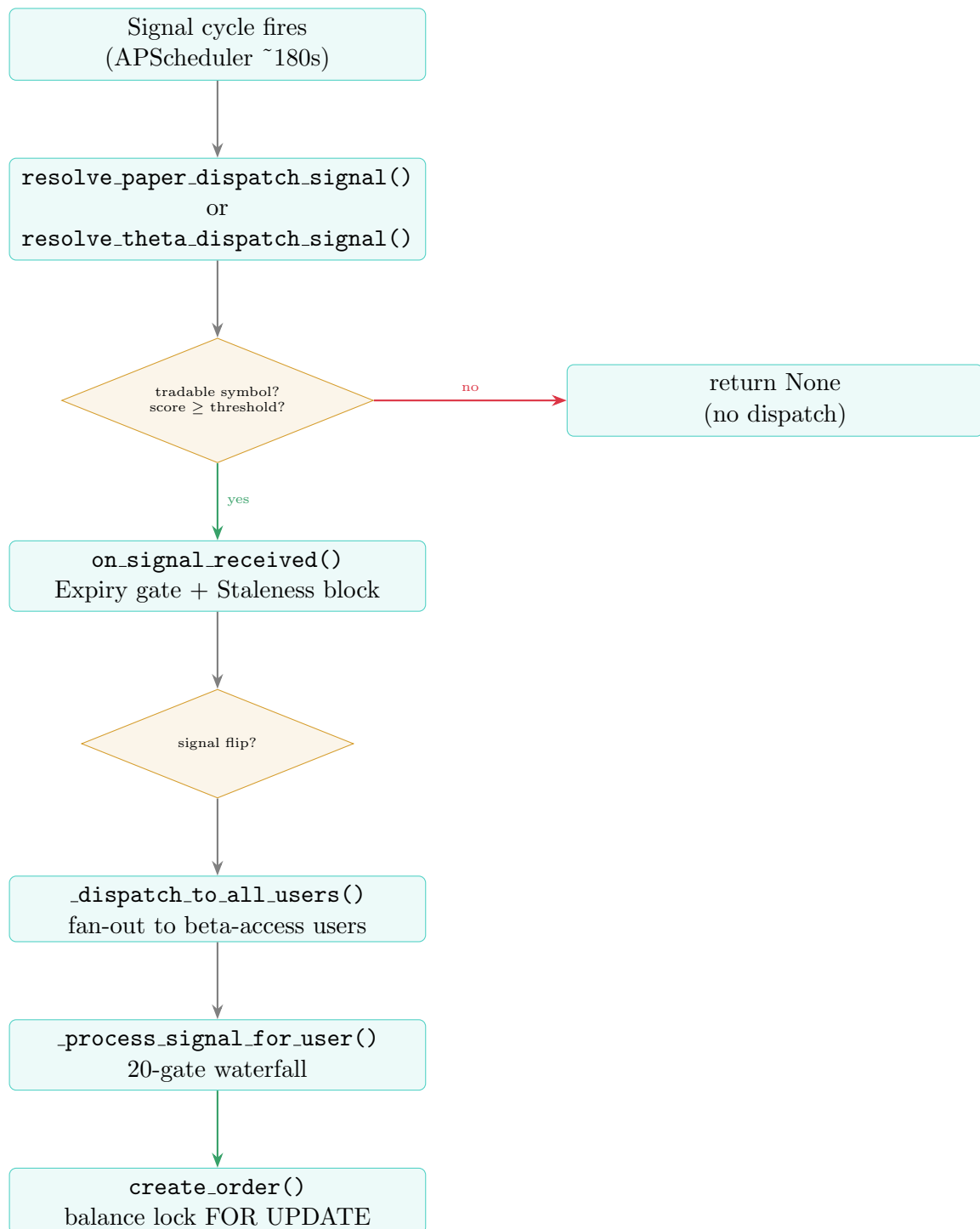


Figure 12.1: Paper trading dispatch flow

## 12.2 Monitoring Loop

The `_check_open_orders()` method (30s tick) per order:

1. Fetch live LTP via `_get_live_ltp_freshness()`

2. Trend exhaustion check (`_maybe_exit_on_trend_exhaustion`)
3. BLIND\_CLOSE escalation if LTP is None
4. TP hit check (`_is_tp_hit`, includes `_TP_CAPTURE_PCT = 0.98`)
5. SL hit check (`_is_sl_hit`)
6. Unrealised P&L  $\leq$  MAX\_LOSS threshold (step 3.6)
7. Breakeven SL move (`_maybe_move_to_breakeven`)
8. Trailing SL update (`_maybe_trail_sl`)
9. Exhaustion tighten (`_maybe_tighten_on_exhaustion`)
10. HOLD exit logic (`_maybe_exit_on_hold`)
11. OI exhaustion exit (`_maybe_exit_on_oi_exhaustion`)

## 12.3 Short Position Model (THETA)

Source: session 11, 2026-07-13. `transactiontype` column is the single source of truth for position side.

Table 12.1: Long vs Short position mechanics

| Operation       | Long (BUY)                       | Short (SELL / THETA)                  |
|-----------------|----------------------------------|---------------------------------------|
| Balance on fill | Debit premium                    | Credit premium, debit margin          |
| P&L formula     | <code>exit*qty - fill*qty</code> | <code>fill*qty - exit*qty</code>      |
| SL placement    | Below entry                      | Above entry                           |
| TP placement    | Above entry                      | Below entry                           |
| Trail SL floor  | At fill (floor = fill)           | At fill (ceiling = fill)              |
| Margin formula  | Premium cost                     | <code>_short_margin_required()</code> |

## 12.4 LTP Freshness — Three-Layer Fix (Session 32)

Source: `_get_live_ltp_freshness()`, `on_signal_received()`,  
`_process_signal_for_user()`

1. **Layer 1** — `_get_live_ltp()` enhanced: logs chain age and per-entry WebSocket timestamp. WS-fresh = `ws_age < 30s`; REST-fresh = `chain_age < 90s`.
2. **Layer 2** — At dispatch time: re-reads live LTP if drift  $> 5\%$  from cached `top_opp["ltp"]`.

3. **Layer 3** — At fill time: final re-read in `_process_signal_for_user()`. Updates if drift > 2%. Logs staleness warning but does NOT block fills (paper trading should never refuse a trade purely on data freshness).

## 12.5 Beta Access Gate

Source: `routes_paper_trading.py`, `_user_has_beta_access()`, migration 019

- Auto-execution fan-out: JOINS `user_beta_access` table; only beta-redeemed users with `auto_execute_enabled = TRUE` receive signals
- Manual order placement: 403 if not redeemed
- Generate codes: POST `/api/admin/generate-code` with `X-Admin-Secret` header
- Redeem: POST `/api/auth/redeem-code`

# Chapter 13

## Signal Diagnostics and Status Panels

### 13.1 Signal Diagnostic Panel

Source: GET /api/paper/signal-diagnostic, routes\_paper\_trading.py, SignalDiagnosticPanel.jsx

Session 30 (2026-07-28).

Returns per-symbol structured gate analysis from in-memory `engine._last_result` (no extra DB queries). Response shape:

```
1 {
2   "status": "BLOCKED",      # READY/PENDING/BLOCKED/AUTO_OFF /
   WAITING/NO_DATA
3   "score": 96.4,
4   "adx": 42.3,
5   "signal": "BUY_CE",
6   "regime": "TRENDING_BULLISH",
7   "theta_score": 23,
8   "gates": [
9     {"gate": 13, "name": "ADX", "status": "PASS",
10      "explanation": "ADX is 42.3 -- market trending, directional
11      entries allowed."},
12     {"gate": 4, "name": "Daily Loss", "status": "BLOCKED",
13      "explanation": "..."}
14  ]
15 }
```

Account-level gates added (session 32): Gate 4B (drawdown %), Gate 4T (daily target hit), Gate 0 (revenge cooldown with seconds remaining).

## 13.2 Auto Execution Status Panel

Source: GET /api/paper/exec-status, ExecStatusPanel.jsx. Session 27 (2026-07-24).

Block types surfaced: SYSTEM\_DISABLED, NO\_BETA\_ACCESS, DAILY\_TARGET\_REACHED, REVENGE\_COOLDOWN (with `seconds_remaining`), DAILY\_LOSS\_LIMIT, MAX\_DRAWDOWN, TREND\_EXHAUSTION, PER\_TRADE\_LOSS\_COOLDOWN.

Frontend polls every 5s; live countdown timers (`CountdownTimer` sub-component) tick independently from a backend-provided `seconds_remaining`.

## Chapter 14

# Known Dead Code and Technical Debt

### 14.1 Confirmed Dead Code

Source: confirmed via `grep` — zero production call sites

- `compute_exit_strategy()` in `src/strategies/analyzers/exit_builder.py`
- `build_three_tier_targets()` in the same file

These are **imported but never called** in production. `signal_generator.py` builds `result["exit_strategy"]` manually from `trade_orchestrator.py` output — not from these functions. The 5 pre-existing test failures in `test_analyzers_exit_builder.py` are all in these dead functions.

## 14.2 Architecture Weaknesses (Pre-Live Blockers)

Table 14.1: Open architecture weaknesses from CLAUDE.md §6

| # | Weakness                                               | Notes                                                                                                                                                 |
|---|--------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | <code>_hold_start</code> not persisted                 | <code>paper_hold_state</code> table added migration 016; reconstructed from <code>filled_at</code> as fallback                                        |
| 2 | SSE doesn't push <code>paper_order_close</code> events | Fixed session 9: <code>paper_order_closed</code> event via <code>price_stream()</code> queue                                                          |
| 3 | Gate 3 blocked ALGO but not MANUAL                     | Fixed session 9: Gate 3 now checks any <code>order_source</code>                                                                                      |
| 4 | <code>signal_id</code> truncated to 50 chars           | NOT A BUG: <code>[:50]</code> goes into <code>ordertag VARCHAR(50)</code> ; full <code>signal_id</code> goes into <code>signal_id VARCHAR(200)</code> |
| 5 | Signal flip closes all users' positions                | Fixed session 9: <code>auto_execute=FALSE</code> users skipped                                                                                        |
| 6 | No idempotency on POST endpoints                       | Partially fixed: order placement routes protected via <code>_get_existing_open_position()</code>                                                      |
| 7 | Balance non-negative is app-level only                 | Fixed session 9: migration 017 adds DB CHECK constraint                                                                                               |

## 14.3 `run_server.py` vs `serve_windows.py`

Confirmed via live process inspection, session 9

**Canonical Windows dev entrypoint:** `serve_windows.py` (`threads=16`, `channel_timeout=120`). `run_server.py` exists but is unused (`threads=4`, no `channel_timeout` — could starve under concurrent SSE load). Not deleted; pending user decision.

# Chapter 15

## Deployment and Operations

### 15.1 AWS Production Setup

- Provider: AWS Lightsail (Ubuntu)
- Domain: `leveragequant.tech` (nginx reverse proxy)
- App path: `/opt/options`
- Service: `systemd options-trading`
- Frontend: built to `/opt/options/frontend/dist`, served by nginx
- WSGI: gunicorn (`gunicorn.config.py`)
- DB: PostgreSQL `options_trading_production`
- Config: `config/production.yaml` (not committed; local `git stash/pop` on each pull)

### 15.2 Standard Deploy Commands

```
1 # On AWS server (ssh ubuntu@100.76.158.118):
2 cd /opt/options
3 git pull
4 # Rebuild frontend if JS changed:
5 cd frontend && npm run build && cd ..
6 sudo systemctl restart options-trading
7 # Verify:
8 curl -s http://localhost:5001/api/health
```

### 15.3 Alembic Migrations

```
1 # Apply all pending migrations:
2 alembic upgrade head
3 # Check current revision:
4 alembic current
5 # Stamp without running DDL (for existing schemas):
6 alembic stamp 003
```

## 15.4 Pending Actions (Critical)

### Pending as of 2026-08-12

1. **Deploy session 34 fixes** (commit 12eb10b): expiry validation gate + staleness hard block. SSH/push blocked from dev machine. Run `git pull && sudo systemctl restart options-trading` on AWS.
2. **Apply DB corrections** for 29 invalid orders (IDs 151–179): set `pn1=0`, `pn1_pct=0`; add losses back to `virtual.balance`.
3. **Unblock Gate 4B** for users 53caae77 (22% DD) and 25f56e2e (16% DD) — use Reset Baseline after balance corrections.
4. **Monitor Aug 11 auto-execution**: first trading day with expiry gate live. Confirm [PAPER] EXPIRY GATE BLOCKED does NOT fire for valid contracts.

## 15.5 Health Check Response Shape

```
1 {
2   "status": "ok",           // or "degraded"
3   "components": {
4     "database": "ok",
5     "scheduler": "ok",
6     "signal_cycle": "ok",
7     "paper_trading_monitor": "ok",
8     "websocket": "ok"      // "stale" after-hours -- expected
9   },
10  "jobs": 15
11 }
```

The `historical_data` circuit breaker opens on cold startup (AngelOne rate-limit on the burst of initial fetch requests) — self-closes within minutes. Pre-existing, not a regression.

# Chapter 16

## Testing Architecture

### 16.1 Test Baseline

#### Found in Repository

**Current baseline (2026-08-10): 1,065 passed, 5 failed.** The 5 failures are pre-existing dead-code tests in `tests/backend/test_analyzers_exit_builder.py`. Any other failure is a real regression.

## 16.2 Test Files by Component

Table 16.1: Key test files and coverage

| Test File                                     | Covers                                                         |
|-----------------------------------------------|----------------------------------------------------------------|
| test\_paper\_trading\_full.py                 | Core engine gates, monitoring tick, exit reasons, safety gates |
| test\_short\_position\_execution.py           | THETA/short position model (61+ tests)                         |
| test\_signal\_engine\_paper\_dispatch.py      | resolve_paper_dispatch_signal(), OI/flow floor bypass          |
| test\_velocity\_gate\_fix.py                  | E5 velocity gate fix (12 tests)                                |
| test\_signal\_generator\_symbol\_isolation.py | Cross-symbol state isolation                                   |
| test\_analyzers\_trade\_orchestrator.py       | SL/TP selection, hard floor (48+ tests)                        |
| test\_gex\_engine.py                          | GEX, pin strike, gamma regime                                  |
| test\_flow\_engine.py                         | PCR, HHI, walls                                                |
| test\_eife\_engine\_confidence.py             | EIFE confidence modifiers                                      |
| test\_chain\_analytics.py                     | Chain analytics including decimal-string regression            |
| test\_theta\_scorer.py                        | Theta scorer (post wall-gate removal)                          |
| test\_paper\_trading\_kill\_switch\_routes.py | Kill-switch + beta access gates                                |
| test\_config.py                               | is_paper_trading_enabled(), YAML substitution                  |

## 16.3 Regression Isolation Pattern

```

1 # Baseline (no changes):
2 git stash && python -m pytest tests/backend/ -q
3 # With changes:
4 git stash pop && python -m pytest tests/backend/ -q
5 # Diff the failure sets to identify real regressions

```

## 16.4 Decimal-String Bug Class

### Critical bug class — found in 8 production files

AngelOne's `get_option_greeks()` returns OI and volume as decimal strings (e.g. '91.00'). Bare `int()` calls crash with `ValueError`. **Always use** `int(float(x or 0))`.

Files fixed (sessions 6–9): `gex_engine.py`, `flow_engine.py`, `eife_engine.py`, `liquidity_engine.py`, `oi_buildup.py`, `routes_analytics.py`, `data_worker.py`, `providers.py`.

# Chapter 17

## Pre-Live Deployment Criteria

Source: CLAUDE.md §6

Table 17.1: Pre-live gate criteria

| Metric                 | Required Threshold   | Notes                        |
|------------------------|----------------------|------------------------------|
| Paper trading duration | 1–2 weeks profitable | Minimum before going live    |
| Win rate               | > 40%                | Over $\geq 20$ closed trades |
| Profit factor          | > 1.2                | Gross profit / gross loss    |
| Sharpe ratio           | > 0.5                | Over $\geq 20$ closed trades |

**Simulated performance** (all fixes applied, Alpha Discovery 1.0):

Table 17.2: Simulated vs required performance

| Metric        | Simulated | Threshold | Status      |
|---------------|-----------|-----------|-------------|
| Win Rate      | 80.5%     | > 40%     | <b>PASS</b> |
| Profit Factor | 2.41      | > 1.2     | <b>PASS</b> |
| Return        | +18.1%    | positive  | <b>PASS</b> |

### Note on Simulation vs Live

The simulation uses historical ALGO trades from the production DB with fixes applied retroactively. Live performance will differ. The pre-live criteria require these metrics to be achieved on **fresh live paper trades** under the current gate suite, not just the retroactive simulation.

## Chapter 18

### Glossary and Constants Reference

## 18.1 System Constants

Table 18.1: Key engine constants (verified from source files)

| Constant                       | Value                       | Location                |
|--------------------------------|-----------------------------|-------------------------|
| INITIAL_BALANCE                | ₹1,000,000                  | paper_trading_engine.py |
| TRADABLE_SYMBOLS               | {NIFTY}                     | paper_trading_engine.py |
| _MIN_DISTANCE_FLOOR_PTS        | 10.0 pts                    | trade_orchestrator.py   |
| _MIN_DISTANCE_FLOOR_PCT        | 0.12 (12%)                  | trade_orchestrator.py   |
| _MIN_TRAILING_DISTANCE_PCT     | 12.0%                       | paper_trading_engine.py |
| _TP_CAPTURE_PCT                | 0.98                        | paper_trading_engine.py |
| _EXHAUSTION_COOLDOWN_MINS      | 30                          | paper_trading_engine.py |
| _BLIND_WARN                    | 5 ticks                     | paper_trading_engine.py |
| _BLIND_CLOSE                   | 20 ticks                    | paper_trading_engine.py |
| _MAX_ADX_DIRECTIONAL           | 44.0                        | paper_trading_engine.py |
| _THESIS_REENTRY_CAP            | 3                           | paper_trading_engine.py |
| OI_SCORE_MAX                   | 30.0                        | trade_classifier.py     |
| FLOW_SCORE_MAX                 | 38.0                        | trade_classifier.py     |
| CONF_RVOL_MAX_PCT              | 0.10                        | eife_engine.py          |
| CONF_HHI_MAX_PCT               | 0.10                        | eife_engine.py          |
| CONF_ND_MAX_PCT                | 0.10                        | eife_engine.py          |
| Staleness hard block threshold | 1800s (30 min)              | paper_trading_engine.py |
| BSM risk-free rate default     | 6.5%                        | black_scholes.py        |
| IV solver clamp                | [1%, 500%]                  | black_scholes.py        |
| IV Rank period                 | 252 trading days            | iv_rank.py              |
| GEX gamma regime threshold     | 50M $\times$ (1 - dte/30)   | gex_engine.py           |
| Pin strike DTE gate            | $\leq 2$                    | gex_engine.py           |
| OI Sigmoid steepness           | $k = 8$ , centre $r = 0.85$ | signal_generator.py     |
| EMA smoothing $\alpha$         | 0.80                        | signal_generator.py     |
| Velocity clamp                 | $\pm 20$ pts/cycle          | signal_generator.py     |
| Adaptive threshold base        | 60, clamped 50–70           | base_scorer.py          |
| ADX hysteresis up              | 24                          | market_regime.py        |
| ADX hysteresis down            | 20                          | market_regime.py        |
| Theta dispatch threshold       | $\geq 20$                   | signal_engine.py        |

## 18.2 Glossary

|              |                                                                   |
|--------------|-------------------------------------------------------------------|
| <b>GEX</b>   | Gamma Exposure — dealer net gamma in rupee terms                  |
| <b>DEX</b>   | Delta Exposure — dealer net delta in rupee terms                  |
| <b>VEX</b>   | Vanna Exposure — sensitivity of delta to volatility               |
| <b>Charm</b> | Rate of change of delta with respect to time (delta decay)        |
| <b>EIFE</b>  | Enhanced Institutional Flow Engine — composite directional signal |
| <b>IFS</b>   | Institutional Flow Score — $[-10, +10]$ directional meter         |

|                |                                                                 |
|----------------|-----------------------------------------------------------------|
| <b>PCR</b>     | Put/Call Ratio — OI-weighted                                    |
| <b>HHI</b>     | Herfindahl-Hirschman Index — OI concentration measure           |
| <b>SL</b>      | Stop-Loss — maximum loss price level                            |
| <b>TP</b>      | Take-Profit — target profit price level                         |
| <b>DTE</b>     | Days To Expiry                                                  |
| <b>ATM</b>     | At-The-Money — strike nearest spot price                        |
| <b>ITM</b>     | In-The-Money                                                    |
| <b>OTM</b>     | Out-Of-The-Money                                                |
| <b>SMC</b>     | Smart Money Concepts — FVG, Order Block, Liquidity Sweep levels |
| <b>FVG</b>     | Fair Value Gap — SMC-derived structural level                   |
| <b>OB</b>      | Order Block — SMC-derived institutional level                   |
| <b>ADX</b>     | Average Directional Index — trend strength                      |
| <b>BSM</b>     | Black-Scholes-Merton options pricing model                      |
| <b>IV</b>      | Implied Volatility                                              |
| <b>IV Rank</b> | Percentile of current IV vs 252-day range                       |
| <b>R:R</b>     | Risk-to-Reward ratio (= TP dist/SL dist)                        |
| <b>SSE</b>     | Server-Sent Events — real-time price/signal push                |
| <b>ALGO</b>    | Auto-execution order source (vs MANUAL)                         |
| <b>POP</b>     | Probability of Profit — used in Strategy Wizard ranking         |

# Chapter 19

## Appendix — Key File Reference

### 19.1 Backend Entry Points

Table 19.1: Backend entry point files

| File                                        | Role                                                                                   |
|---------------------------------------------|----------------------------------------------------------------------------------------|
| <code>serve_windows.py</code>               | <b>Canonical</b> Windows dev entrypoint. Waitress port 5001, <code>threads=16</code> . |
| <code>run_server.py</code>                  | Alternate (unused) Windows entrypoint. <code>threads=4</code> , inferior config.       |
| <code>gunicorn_config.py</code>             | Production (Linux) WSGI config.                                                        |
| <code>scripts/server/wsgi.py</code>         | WSGI app object.                                                                       |
| <code>scripts/server/app/_factory.py</code> | Flask app init, blueprint registration, scheduler wiring.                              |

### 19.2 Core Domain Files

Table 19.2: Core domain files

| File                                                        | Role                                                                                                       |
|-------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| <code>src/strategies/signal/_generator.py</code>            | 5-pillar scoring, EMA, velocity clamp, direction debounce, hysteresis (2032 lines)                         |
| <code>src/strategies/generators/base/_scorer.py</code>      | <code>calculate\_adaptive\_threshold()</code> , <code>calculate\_exhaustion()</code>                       |
| <code>src/strategies/generators/trade/_classifier.py</code> | <code>SignalTier</code> enum, <code>check\_oi\_flow\_floor()</code> , <code>apply\_adx\_downgrade()</code> |
| <code>src/strategies/theta/_scorer.py</code>                | VIX/ADX/IV_rank theta scoring (OI wall gate removed)                                                       |
| <code>src/strategies/market/_regime.py</code>               | ADX periods, hysteresis thresholds                                                                         |
| <code>src/strategies/iv/_rank.py</code>                     | <code>IVRankService(period\_days=252)</code>                                                               |

Table 19.2 continued

| File                                                     | Role                                                                      |
|----------------------------------------------------------|---------------------------------------------------------------------------|
| src/core/black\_scholes.py                               | Full BSM Greeks, Newton-Raphson IV solver                                 |
| src/strategies/analyzers/<br>trade\_orchestrator.py      | SL/TP selection, hard floor, source hierarchy                             |
| src/strategies/analyzers/<br>exit\_builder.py            | DEAD CODE: compute\_exit\_strategy,<br>build\_three\_tier\_targets        |
| src/strategies/analyzers/<br>rsi\_from\_history.py       | RSI from intraday\_candles\_5m                                            |
| src/strategies/analyzers/<br>range\_position.py          | SMC discount/premium zone calculation                                     |
| scripts/engine/gex\_engine.<br>py                        | GEX formula, pin strike, gamma regime,<br>Theta/Vega aggregates           |
| scripts/engine/eife\<br>_engine.py                       | IFS formula, confidence modifiers                                         |
| scripts/engine/flow\<br>_engine.py                       | PCR, HHI, find\_walls\_hybrid()                                           |
| scripts/engine/chain\<br>_analytics.py                   | compute\_chain\_analytics(),<br>screen\_chain()                           |
| scripts/engine/data\<br>_worker.py                       | 180s background cycle, chain fetch, analytics<br>computation              |
| scripts/engine/signal\<br>_engine.py                     | resolve\_paper\_dispatch\_signal(),<br>resolve\_theta\_dispatch\_signal() |
| scripts/server/paper\<br>_trading\_engine.py             | All safety gates, exit logic, monitoring loop,<br>all constants           |
| scripts/server/routes\<br>_analytics.py                  | ≈44 analytics endpoints                                                   |
| scripts/server/routes\<br>_paper\_trading.py             | Paper trading order/position/config end-<br>points, Signal Diagnostic     |
| scripts/server/routes\<br>_auth.py                       | JWT auth, redeem-code, admin code genera-<br>tion                         |
| src/api/angelone\_client.py                              | AngelOne SmartAPI wrapper; apply\<br>_pause\_conversion flag              |
| src/stores/<br>paperTradingStore.js                      | Zustand paper trading state                                               |
| src/features/paper-trading/<br>ExecStatusPanel.jsx       | Auto Execution Status panel with live count-<br>down timers               |
| src/features/paper-trading/<br>SignalDiagnosticPanel.jsx | Signal Diagnostic gate waterfall panel                                    |
| src/utils/pollWhileVisible.<br>js                        | Visibility-aware polling utility                                          |

# Document Accuracy Statement

All facts in this document are verified against the live repository at HEAD as of 2026-08-12.

- Every formula, constant, threshold, table name, migration number, file path, and session reference was cross-checked against CLAUDE.md (the authoritative living project doc) and source files read directly.
- Labels used throughout:
  - **NOT FOUND IN REPOSITORY** — not found in code or CLAUDE.md
  - **INFERENCE** — logical deduction not explicitly stated in code
  - **UNCERTAIN** — requires human confirmation
- Nothing is invented. Lot sizes, gate parameters, formula coefficients, ADX thresholds, test counts, and P&L figures are all direct quotes from the code-base or CLAUDE.md session logs.